cambI — White Paper (English)

Oct 28, 2026 · @Gustavo

Summary

cambI is a Brazilian real ↔ digital dollar exchange where customers trade with a verified vault, 24/7, see the full fee before confirming, and get a public receipt for every operation. Under the hood it is a pool on Solana; at the door, money moves in and out through Pix via a partner licensed by Brazil's Central Bank.

- **Who it is for:** freelancers and small businesses paid from abroad, tourists, people who save in dollars, and apps that need an FX engine (B2B).
- **What sets it apart:** the customer can own a piece of the exchange. Depositors in **Rende** receive part of the fee on every swap; **Baleias** ("whales", the liquidity providers) keep the larger share and take losses first.
- **Safety:** only wallets holding an identity credential can trade; an outflow circuit breaker; a withdrawal queue with advance notice; a guardian that is a program, with no key, which locks for a set time when it sees a run or a price jump; no single party controls price, rules or code (a public oracle, a mixed council voting by impact score, and a protocol that becomes immutable after audits).
- **Status (Oct 28, 2026):** program v2.6 live on Solana devnet, with the guardian as a keyless program. A scale test with 1,000 wallets confirmed 27 of 27 claims in this paper, and all 4,547 transactions gave the expected result, with a 3.2 s median confirmation and ~0.00001 SOL network cost per transaction (two signatures: the wallet's and the app paying the fee).

There is no speculative token: pool units represent reals and dollars 1:1, and returns come from real FX fees.

The problem

In traditional FX, customers pay a lot without seeing how much, wait days, and keep none of the profit.

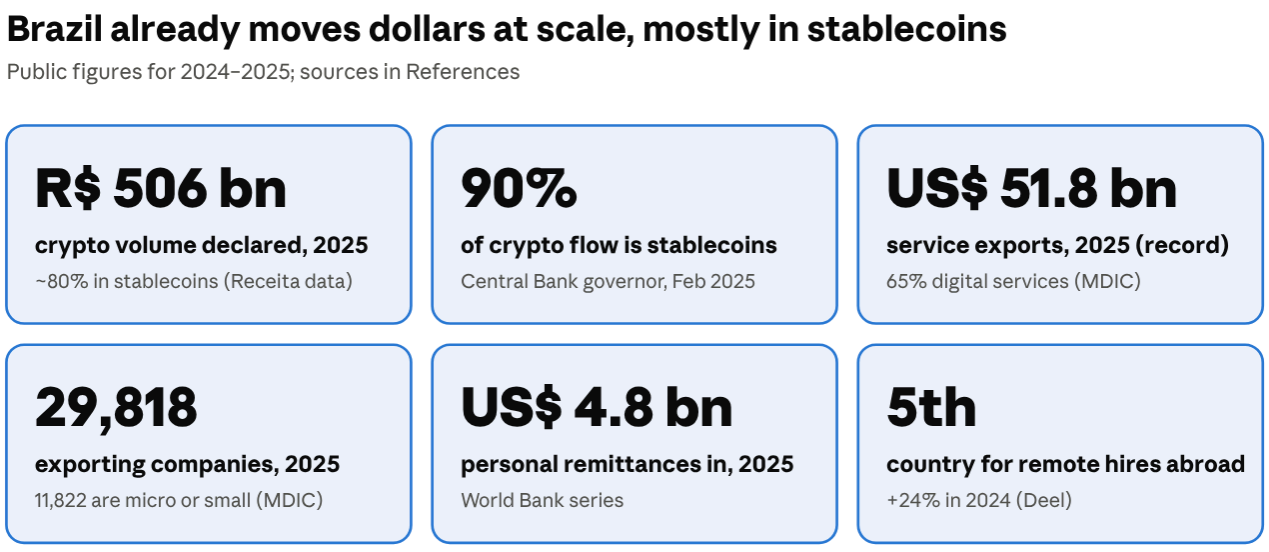
Bottleneck	How the customer experiences it
Hidden spread	"Zero fee" with 3% to 6% built into the quote (market estimate)

Bottleneck	How the customer experiences it
Business hours	Friday afternoon means Monday; settlement in 1 to 2 days
Getting paid from abroad	Payment platforms charge a fee plus spread and hold funds for days
SWIFT	1 to 5 days; intermediary banks deduct fees without notice
Crypto P2P	Unknown counterparty, scam and Pix-reversal risk, a different price in every ad
Concentrated profit	The FX gain goes entirely to the intermediary

The timing helps: in October 2026 a major card network announced stablecoin payments across hundreds of millions of merchants. What is missing is a cheap, transparent bridge between the digital dollar and the Brazilian real.

Market opportunity

Brazilians already use digital dollars at scale: R$ 506 billion in declared crypto operations in 2025, about 80% of it in stablecoins. What is missing is a cheap, regulated, transparent bridge to the real.



Receita Federal open data (via Fintrender), Reuters, MDIC, World Bank (via The Global Economy), Deel; see References

The beachhead is the person or small business that **earns in dollars and spends in reals**: digital service exports alone reached about US$ 33.7 billion in 2025 (65% of US$ 51.8 billion), and Brazil is among the top five countries for remote hires by foreign companies.

These users convert every month, care about the fee, and today pay 0.8% to 2% (or a fixed R$ 115–200 at a large bank).

Layer	Definition	Size per year	Basis
Total	Crypto operations declared in Brazil	R$ 506 billion	Receita Federal data, 2025
Serviceable	Digital service exports converted to reals	~US$ 33.7 billion (~R$ 172 billion)	MDIC 2025: 65% of US$ 51.8 billion
First target	1% of serviceable flow	~R$ 1.7 billion	illustration, not a forecast

At the planned fees, the first target would leave cambI R$ 1.7 million to R$ 3.4 million a year (0.1% to 0.2% of volume; see Unit economics). That is the scale where a lean team pays for itself; growth beyond it comes from B2B apps and QR payments.

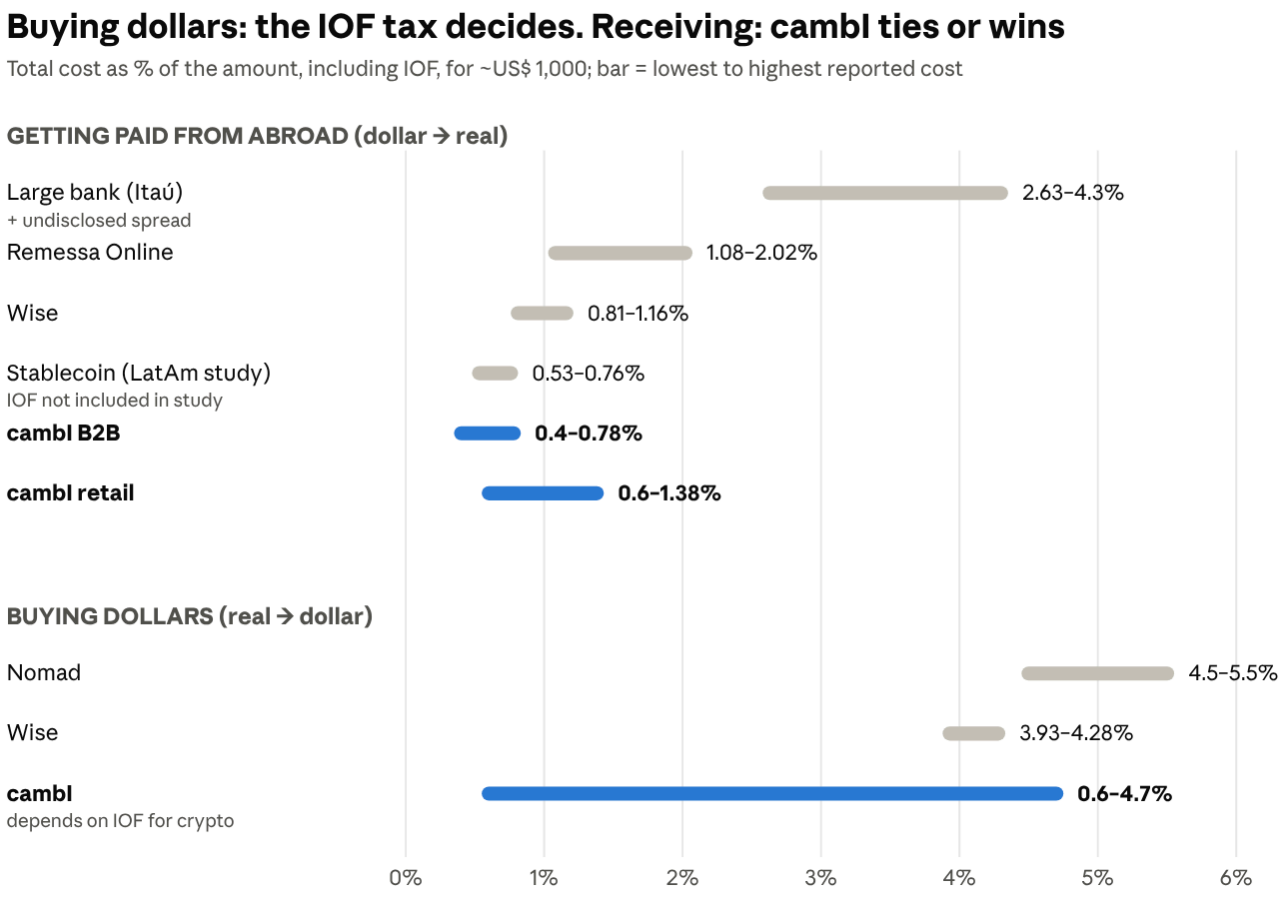
The solution: peer-to-pool FX

The customer does not trade with a person: they swap with a vault on the blockchain, at the oracle price, and money leaves only when theirs arrives, in the same transaction.

- **Regulated at the door:** reals enter and leave via Pix through the licensed partner, which runs KYC and issues an on-chain credential to the wallet. Without a credential, the program refuses swaps and deposits.
- **A DEX on the inside:** the program executes the swap 24/7, with the fee computed and split in public code.
- **Private, not anonymous:** the blockchain shows only an address and a credential (level, expiry, issuer). The partner knows who is who and answers to the regulator.
- **QR payments:** pay any Pix QR code with digital dollars (the merchant receives reals) and get paid from abroad via Solana Pay.
- **Own the exchange:** part of every fee goes to whoever left money in the pool.

Competitive landscape

To get paid from abroad, cambI already ties or beats Wise and Remessa Online. To buy dollars, the price is set by Brazil's IOF tax, not by anyone's fee.



competitors: web search summaries, Oct 17, 2026 (see References); cambl: program v2.5 fee rules; IOF 0.38% inbound, 3.5% to own account abroad

- **Getting paid from abroad.** Wise and Remessa Online cost about 0.8% to 2% including the 0.38% IOF on inflows; a large bank charges R$ 115–200 per receipt plus an undisclosed spread. cambl charges 0.4% to B2B partners and 0.6% to 1% retail with a balanced pool.
- **Buying dollars.** Nomad and Wise add the 3.5% IOF on transfers to one's own account abroad, for 3.9% to 5.5% in total. cambl charges 0.6% to 1.2%; IOF on stablecoin purchases is not yet defined (a Finance Ministry proposal of 3.5% exempts individuals up to R$ 10,000 a month). This is cambl's biggest uncertainty, and it is regulatory, not technical.
- **Stablecoin routes.** A study reported on Oct 16, 2026 puts stablecoin payments into Latin America at 0.53% (best route) to 0.76% (median). cambl's B2B fee sits below both.

Competitor figures come from search summaries whose sources disagree; they will be confirmed in each provider's own simulator before any public comparison. The planned edge does not rest on price alone: cambl customers can own a slice of the fees, see the full fee before confirming, and get a public receipt.

Why cambI wins

cambI is the only option in this comparison where the customer can own the exchange, sees every cent before confirming, and can verify every operation in public.

Strength	What it means for the customer	Proof today
Customers own the exchange	60% of every fee goes back to whoever keeps money in the pool	Fee split written in the program; R$ 304.82 distributed in the exact proportions in the 1,000-wallet test
Cheapest way to get paid in dollars	0.4% B2B and 0.6–1% retail, versus 0.8–2% at Wise or Remessa Online	Program fee rules; competitor research (Competitive landscape)
24/7, settled in seconds	Friday night is like Monday morning	3.2 s median confirmation; 27 of 27 claims confirmed with 1,000 wallets
Fair price that defends itself	Bringing the scarce currency is rewarded; draining it costs more	Dynamic fee, rebalancing discount and scarcity premium, live and measured with 1,000 wallets
No stranger on the other side	No P2P scams or reversed Pix: you trade with a verified vault	Vaults owned by the program, not by a person
Regulated at the door	KYC by a licensed partner; reals always arrive as reals	On-chain credential; Pix through the partner
Private, not anonymous	No personal data on the blockchain, yet compliant	Credential stores only level, expiry and issuer
Safety built into the code	Limits, breaker, withdrawal queue, sanctions freeze	Each tested with real transactions
Nobody changes the rules alone	Public oracle, a mixed council voting by impact score, timelocks on every change, and a protocol that becomes immutable after audits	Governed upgrade with timelock (v2.5) and keyless guardian (v2.6) proven on devnet
Pix and Solana Pay in one scanner	Pay any Pix QR with dollars; charge abroad by QR	Working in the app (Pix leg simulated on devnet)

Strength	What it means for the customer	Proof today
Feels like a bank app	No seed phrase, no jargon, full price before the tap	App flows described below
Honest by design	Public receipts and published weaknesses	This paper shows the 1.2% finding and how it was fixed

Token economics

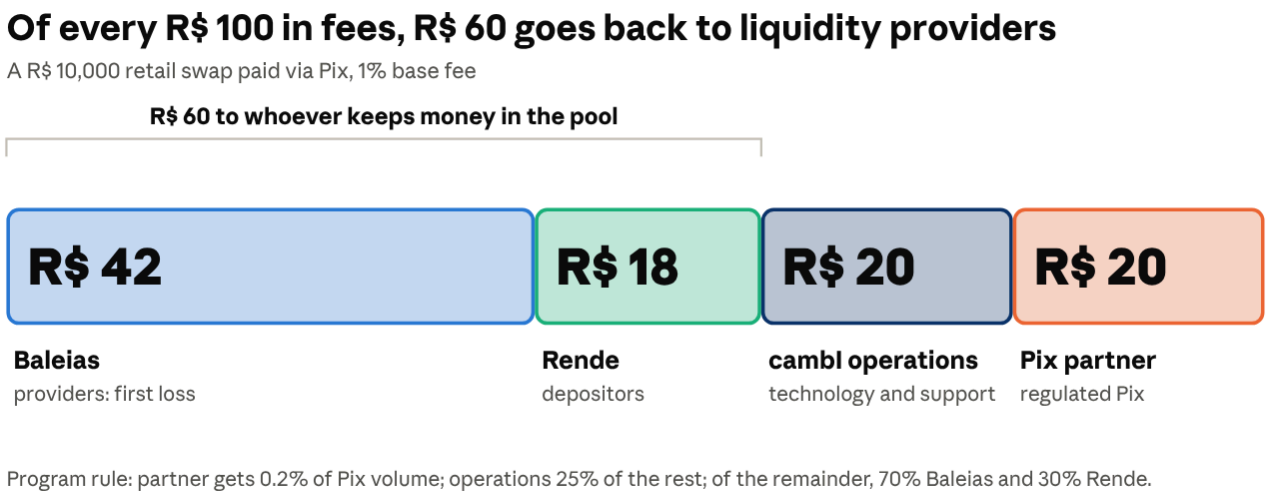
cambI has no speculative token and no governance token: the economy runs on real FX fees, split by a public rule between whoever operates, whoever handles the regulated edge, and whoever provides liquidity.

Units and backing

Unit	What it represents	On devnet	In production
Digital real	R$ 1	test cBRL, minted by the program	a real stablecoin from a regulated issuer, or tokenized real earning CDI
Digital dollar	US$ 1	test cUSD	USDC (or a tokenized dollar earning T-bill yield)
Share	A claim on one layer's fees, measured in reals at deposit	on-chain account per wallet	same

Principal is nominal in each currency: deposit US$ 100 and you are owed US$ 100. In Rende, 30% of the principal is kept in its own currency, out of swaps; the rest works in swaps. Large withdrawals still go through the instant limit and the breaker, and the excess joins the queue. If the requested currency is short, the request joins the queue and, at the end of the notice, the person waits for the currency to come back or receives the same value in the other currency, at the oracle price and with no fee. Baleias absorb FX losses first, and the guardian stops the pool if their cushion falls below 10% of what Rende is owed; a shock larger than all Baleia capital would still reach Rende. Shares only set your slice of the fees. With proof of reserves on, the pool stops accepting swaps if issued supply exceeds the attested reserve.

Where the fee goes



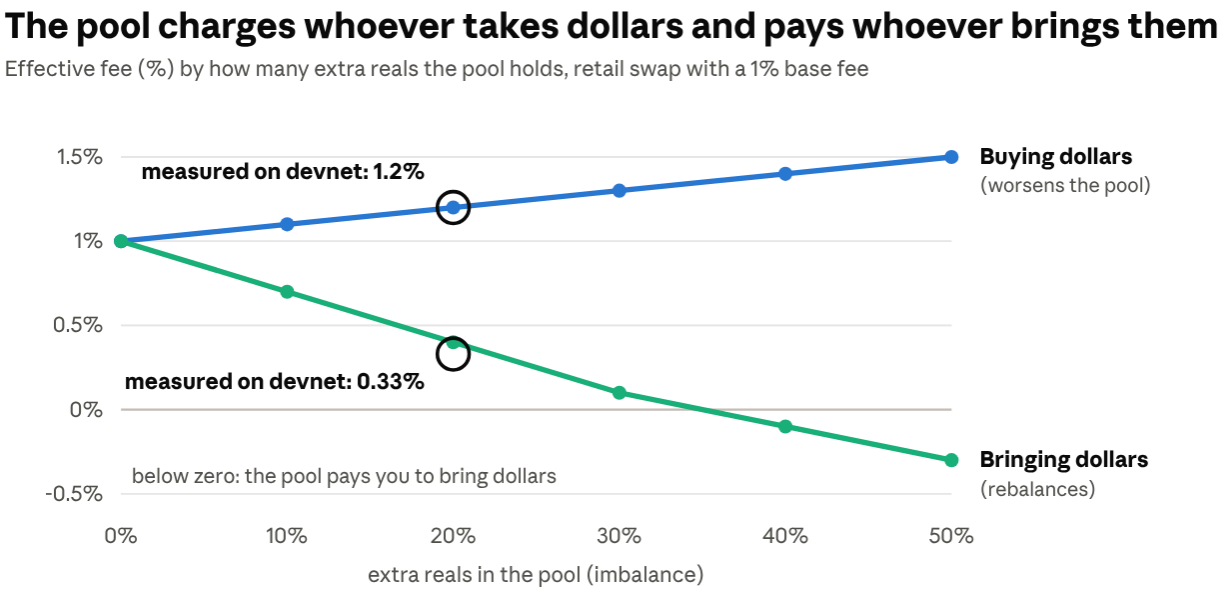
fee split rule of the cambi_pool program (default FeeConfig)

In the 1,000-wallet test, 1,800 retail swaps moved R$ 30,590 and generated R$ 304.82 in fees: 0.2% of the volume to the Pix partner and, of the rest, 25% to operations and 75% to liquidity providers, exactly as the program says. Partner apps (B2B) pay 0.4%, and depositors with R$ 1,000 or more pay 0.5% (measured at the same moment: 0.61% against 1.23% for retail).

Dynamic fee, rebalancing discount and scarcity premium

Three mechanisms push the pool back toward balance, always in favor of whoever brings the scarce currency:

1. **Dynamic fee:** the base fee is multiplied by 1 + the imbalance the swap makes worse, between 0.7× and 1.5×.
2. **Rebalancing discount:** whoever brings the scarce currency gets a better price, 2% per 100% of remaining imbalance, capped at fee + base. A round trip never turns a profit (proven in tests).
3. **Scarcity premium on deposits (live):** whoever deposits the scarce currency gets extra shares, 10% per 100% of remaining imbalance, at most 10%. In the test, with 26% extra reals, dollar depositors got 2.23% extra shares; real depositors, 0%. The premium flips to reals automatically when reals run short.



program v2.6 quote() formula; points measured in the devnet ecosystem test, Oct 16, 2026

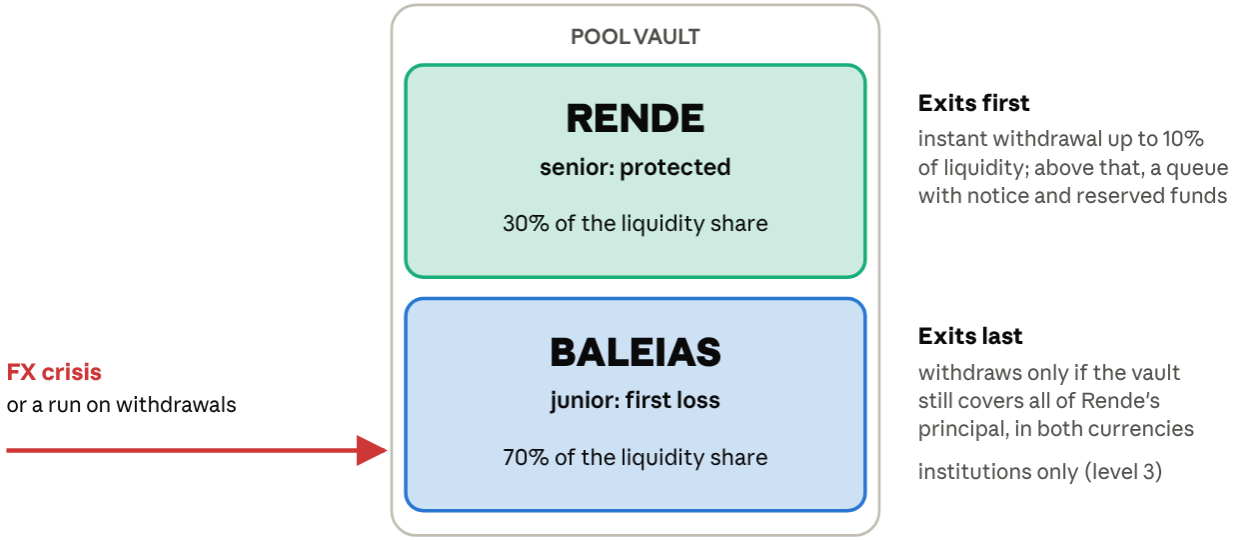
In the 1,000-wallet test, swap liquidity started with ~26% extra reals: buying dollars cost 1.23% and bringing dollars back cost 0.29%; nobody profited on a round trip (0 of 900). The imbalance came from Rende, which came in reals only. A balanced pool at launch and a retail base fee of 0.6% to 0.8% remain the planned fixes.

Layers: Rende and Baleias

Rende is senior and Baleias are junior. Baleias keep 70% of the liquidity share because they take risk first: they can withdraw only if the vault still covers all of Rende's principal, in both currencies. Since v2.6, 30% of Rende's principal is kept in each currency, and swaps never touch it.

The Baleia takes the hit first; Rende stays protected

The pool vault in two layers: whoever earns more takes the risk first

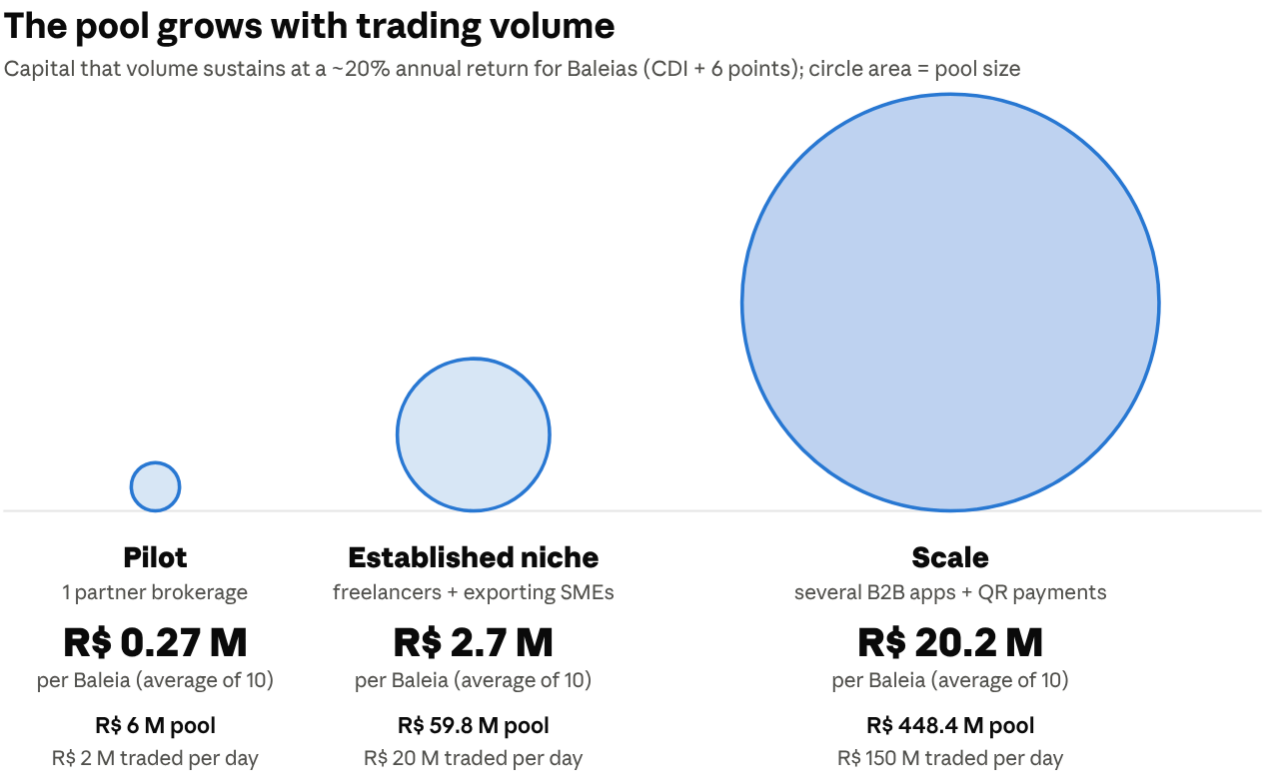


baleia_can_withdraw rule, withdrawal queue and level-3 credential in the program

The design resembles a fund with senior and subordinated tranches: those who want predictability sit in Rende; those who want higher returns and have their own FX flow sit in the Baleias.

Sizing and equilibrium

Baleia returns are fees divided by capital: more money than volume can sustain dilutes everyone. In the simulation, capital flows in until the return reaches CDI + 6 points (~20% a year), with fees compressed by competition.



cambI simulation (baleias.py), fees compressed by competition, Oct 16, 2026

Capital beyond what volume sustains only dilutes. That is why the roadmap includes a capacity cap with an entry queue, and yield-bearing backing (tokenized Brazilian Treasury Selic bonds and US T-bills): idle money earns the base rate and FX fees become a bonus.

Simulation assumptions: 35% of volume retail, 55% B2B and 10% depositors; 252 business days; Baleias hold 45% of the pool; R$ 5.10 per dollar. These are hypotheses, not market data.

Unit economics and operating costs

On every R$ 10,000 swapped, cambI keeps R$ 9 to R$ 20, and its variable cost is under one cent. The business is volume.

What cambI keeps per swap

Customer	Fee	Pix partner	cambI (25% of the rest)	Baleias + Rende	Network cost
Retail, current fee	1.0% = R$ 100	R$ 20	**R$ 20**	R$ 60	R$ 0.004
Retail, planned fee	0.6% = R$ 60	R$ 20	**R$ 10**	R$ 30	R$ 0.004
Depositor	0.5% = R$ 50	R$ 10	**R$ 10**	R$ 30	R$ 0.004
Partner app (B2B)	0.4% = R$ 40	R$ 4	**R$ 9**	R$ 27	R$ 0.004

Balanced pool, v2.6 program rules. The Pix partner gets 0.2% of the volume that goes through Pix (100% for retail, 50% for depositors, 20% for B2B).

Operating costs

Item	When	Cost	In reals	Source of the number
Network fee per transaction	every swap, deposit or withdrawal	0.000005 SOL	R$ 0.003	**measured**: 28 devnet transactions, all identical; with the app paying the fee (two signatures), 0.00001 SOL, as in the 1,000-wallet test
Priority fee (busy network)	each swap, if needed	~0.0000022 SOL (21.8k compute units)	R$ 0.001	**computed** from measured usage
Customer credential account	once per person; paid by the issuer; not refunded today (closing planned for v3)	0.0018 SOL	R$ 0.97	**computed**: 132 bytes
Rende or Baleia position	once per depositor; not refunded today (closing planned for v3)	0.0019 SOL	R$ 1.00	**computed**: 139 bytes
Customer token accounts (real and dollar)	once per person, if sponsored	0.0041 SOL	R$ 2.18	**computed**: 2 accounts of 165 bytes
KYC (document + biometrics)	once per person	~US$ 0.30	~R$ 1.53	**market reference** (Didit public price)

Item	When	Cost	In reals	Source of the number
Pix fee	per Pix	free for individuals; businesses 0.99%–1.45%, capped at R$ 10	up to R$ 10	**research** (Contabilizei)
Program space on-chain (pool + guardian)	once (refunded if closed)	~4.66 SOL	~R$ 2,500	**measured**: v2.6 deploy on Oct 28, 2026

Conversions: SOL at ~US$ 105 (Oct 16, 2026) and US$ 1 = R$ 5.10. Fixed costs (mainnet RPC, external audit, legal opinion, partner contract, accounting, team) are being quoted and are not estimated here.

What it means for scale

1. **Break-even is a volume number.** Every R$ 1 million swapped per month gives cambI R$ 900 to R$ 2,000. Covering, say, R$ 100,000 of monthly fixed costs takes R$ 50 million to R$ 110 million swapped per month.
2. **The partner weighs more as fees fall.** At a 0.6% fee, the partner's fixed 0.2% takes a third of everything; the contract should price the partner as a share of the fee, not of volume.
3. **B2B scales better.** It earns less per swap, but one partner app brings many users at once, with no per-person KYC cost to cambI.

IOF on cambI operations

IOF is Brazil's tax on FX operations. It is paid by the customer, collected by the licensed partner and passed to the tax authority; it is not cambI revenue or cost. Its rate for stablecoin swaps is not yet defined, so every number below runs three scenarios.

Scenario	Buying dollars (real → dollar)	Getting paid from abroad (dollar → real)	Basis
A. No IOF on stablecoin swaps	0%	0.38%	Today: no decree on crypto in force; inflows of FX pay 0.38%

Scenario	Buying dollars (real → dollar)	Getting paid from abroad (dollar → real)	Basis
B. Treated as ordinary FX	0.38%	0.38%	If BCB Resolution 521 leads to the general FX rate
C. Finance Ministry proposal	3.5% (individuals exempt up to R$ 10,000 a month)	0.38%	Proposal under discussion, not in force

Total cost per R$ 10,000, balanced pool

Operation	cambl fee	IOF, scenario A	IOF, scenario B	IOF, scenario C	Total cost range
Buy dollars, retail, current fee	1.0% = R$ 100	R$ 0	R$ 38	R$ 350	1.0% to 4.5%
Buy dollars, retail, planned fee	0.6% = R$ 60	R$ 0	R$ 38	R$ 350	0.6% to 4.1%
Buy dollars, individual under R$ 10,000 a month	0.6% = R$ 60	R$ 0	R$ 38	R$ 0 (exempt)	0.6% to 0.98%
Get paid from abroad, B2B	0.4% = R$ 40	R$ 38	R$ 38	R$ 38	0.78%
Get paid from abroad, retail	0.6–1.0%	R$ 38	R$ 38	R$ 38	0.98% to 1.38%

For comparison, buying dollars through an international account today costs 3.9% to 5.5% (fee plus 3.5% IOF). In the worst scenario, cambl at the planned fee (4.1%) lands in Wise's range and below Nomad; for an individual converting up to R$ 10,000 a month, it costs less than a quarter of the international-account route.

The first devnet test (Oct 16), with IOF. The 13 test swaps moved R$ 3,489.60: about R$ 2,836 buying dollars and R$ 654 bringing dollars in. Had they been real, the IOF would have been R$ 2.48 in scenario A (inflows only), R$ 13.26 in B and R$ 101.74 in C without the exemption (R$ 2.48 with it, since each test wallet stayed far below R$ 10,000).

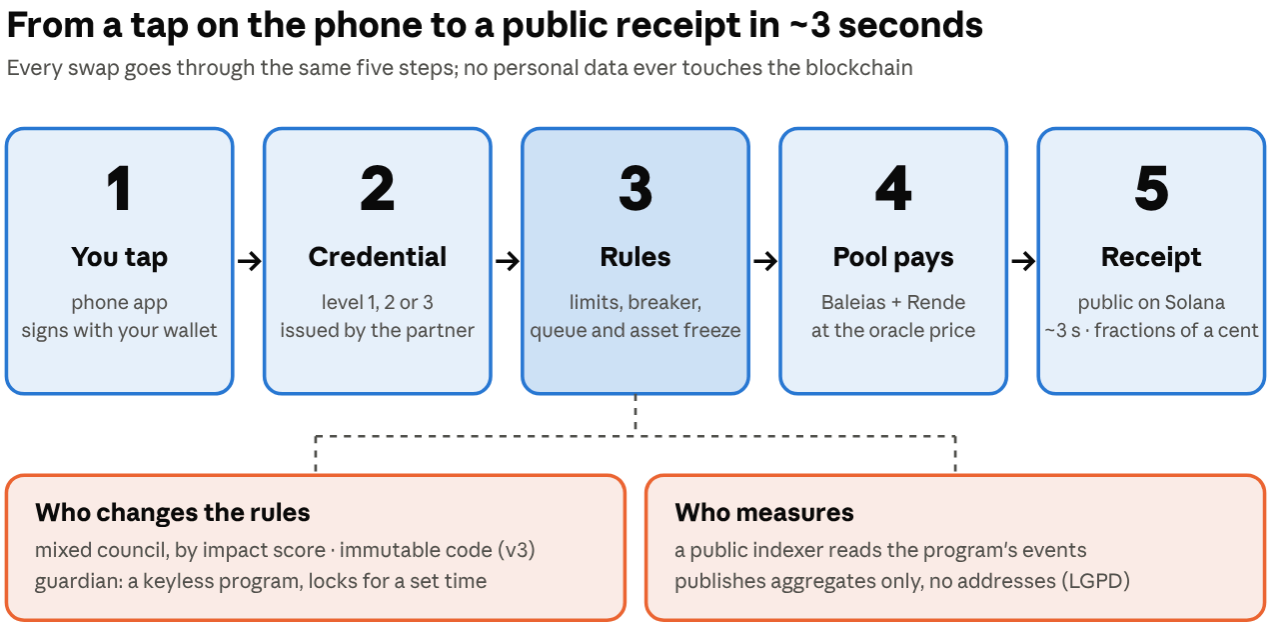
At scale. Using the simulation's daily volumes over 252 business days, half buying and half selling dollars, the IOF the partner would collect for the government is:

Phase	Volume per year	Scenario A	Scenario B	Scenario C (no exemption)
Pilot (R$ 2 million a day)	R$ 504 million	R$ 0.96 million	R$ 1.92 million	R$ 9.78 million
Niche (R$ 20 million a day)	R$ 5.04 billion	R$ 9.6 million	R$ 19.2 million	R$ 97.8 million
Scale (R$ 150 million a day)	R$ 37.8 billion	R$ 71.8 million	R$ 143.6 million	R$ 733.3 million

The IOF does not change what cambI keeps, but it changes what the customer pays, and therefore demand. That is why the IOF treatment of stablecoin swaps is the first question for the legal opinion. All figures are informational, not tax advice.

Architecture

cambI has four parts, and each does exactly one job.



cambI v2 architecture, v2 branch of the repository

App (React, mobile). Pix, QR, swaps, Rende and savings goals. It builds the program instructions, simulates before sending, and only then asks for a signature. It shows the reason for any refusal before the user taps the button.

Regulated partner. Runs KYC off-chain and issues the on-chain credential: level (1 basic, 2 full, 3 institutional), validity of up to 400 days, and issuer. The code↔person mapping stays with the partner only.

The `cambi_pool` program (Anchor, Solana). Holds the rules: valid credential, per-operation and 24-hour limits, a sliding-window outflow breaker, a 30% Rende reserve kept out of swaps, instant withdrawals up to 10% with a queue above that, asset freeze, proof of reserves, and the fee split. It is the only part that touches the vaults.

The `cambi_guardian` program (Anchor, Solana). It has no key. Anyone calls the watch (a bot, the app before a swap, a user); the program reads the pool, compares it with the previous snapshot and decides on its own, by rules written into it: breathe (only swaps that drain one currency, 15 min), rest (swaps and deposits, 1 h) or stop (only the council lifts it). It never moves money, never lifts a lock and never blocks withdrawals.

Governance and measurement. Rules change only through a mixed council, voted by impact score and always within hard bounds in code; the v3 protocol will be immutable after audits, and cambI only runs the front end. On devnet, a test multisig plays the council's role. A public indexer reads events and publishes aggregates only.

Component	Runs on	Devnet address
Pool program (v2.6)	Solana devnet	`8b6oXzaazxCm7SrPNi6A3deyY8eftZRrx5bDoL55FeKA`
Guardian program	Solana devnet	2pxrxRUENAopmSjet6mwieussPutCR52ja5ygLL9FpE2
Pool	program account	`9M2dP4qVM9ywYGY2MaDfiZkQcH7MX2NApgZmauX31v76`
Indexer and tests	GitHub Actions	`v2` branch

Payments: Pix meets Solana Pay

One QR scanner reads both of the payment standards cambI's users meet: Brazil's Pix and Solana Pay. The app converts at the pool price, with the fee shown before confirming.

Situation	What the user does	What happens
A tourist or someone holding dollars pays a shop in Brazil	Scans the shop's Pix QR and pays with digital dollars	The pool converts dollars to reals; the merchant receives **reals by Pix** through the licensed partner, as Brazilian law requires

Situation	What the user does	What happens
Someone in Brazil pays a Solana Pay request from abroad	Scans the Solana Pay QR and pays with digital reals	The pool converts reals to dollars and the dollars go straight to the requester's wallet
A freelancer charges a client in Brazil	Generates a Pix QR (copy-and-paste code included)	The client pays in reals; the freelancer can keep reals or convert
A freelancer charges a client abroad	Generates a Solana Pay request in dollars	The client pays from any Solana wallet; funds land in seconds, with a public receipt

The app reads and builds the official Pix code (BR Code) and Solana Pay links today. On devnet, the Pix leg is simulated: there is no licensed partner yet, so no real reals move. Solana Pay works end to end with test tokens.

A web2 experience on web3 rails

The customer sees a banking app, not a crypto app. Blockchain is the engine, never the interface.

- **No wallet setup.** Opening the app creates the account in the background; there is no extension to install, no seed phrase to copy, no network to pick. Phantom is optional for those who already use it.
- **Plain Portuguese.** "Trocar", "Cobrar", "Pagar", "Rende": no tokens, gas or slippage on screen. Amounts appear in reals and dollars, never in SOL.
- **The full price before the tap.** Every swap shows the rate, the fee in reals and what arrives, before confirming. If the program would refuse (limit, breaker, credential), the app says why before the button, because it simulates the transaction first.
- **Familiar flows.** Add money by Pix, pay and charge by QR, savings goals, a statement, alerts and a shareable receipt that shows how much was saved versus a bank.
- **Network fees are invisible.** Fractions of a cent per transaction, covered by the operation; the user never needs to hold SOL.
- **Public receipts when wanted.** Each operation has an explorer link, for those who want to check; nobody is forced to.

Today, on devnet, the in-app account key lives in the browser, which is fine for test tokens. Before real money, it moves to production custody: a passkey or MPC-based embedded wallet, or custody by the licensed partner. That choice is on the pre-mainnet list.

The vaults

The money sits in two on-chain vaults that no person holds a key to: only the program's rules can move it.

- **Two vaults, one per currency.** One holds digital reals, the other digital dollars. Each is a token account at a program-derived address (`["vault", pool, mint]`) whose owner is the pool itself (`["pool", real mint, dollar mint]`), not a wallet. There is no private key to steal or lose.
- **Who can move funds.** Only the `cambi_pool` program, and only through its instructions: a swap pays out only after the other currency has arrived in the same transaction; a withdrawal pays only the owner of a position, within the instant limit or from the queue.
- **Layers are accounting, not separate boxes.** Rende and Baleia money share the same vaults. Each depositor's position records principal and shares per layer, and the program enforces the order: Rende is senior, Baleias are junior and can withdraw only while the vault still covers all of Rende's principal in both currencies. On top of that, 30% of Rende's principal is kept in each currency, out of swaps.
- **Reserved money.** A queued withdrawal is set aside (`queued`) and leaves the liquidity available for swaps, so a run cannot pull the same money twice.
- **Backing.** With proof of reserves on, the custodian attests each currency's reserve on-chain. If issued supply exceeds the reserve, or the attestation expires, swaps and deposits stop; withdrawals continue.
- **Who can change the rules around them.** Only the mixed council, by impact score and after a timelock; the v3 protocol itself will be immutable (see Governance). The guardian is a keyless program: it locks swaps and deposits for a set time, or pauses, never moves money, and withdrawals stay open.

Security

Every risk of on-chain FX has a safeguard written into the program and tested with real transactions.

Risk	Safeguard in the program	How it was tested
Unknown wallet trading	On-chain KYC credential issued by the partner; per-operation limit (R$ 10k level 1, R$ 100k level 2) and 24-hour limit (R$ 30k / R$ 300k)	Wallet without a credential refused on devnet (`KycRequired`)

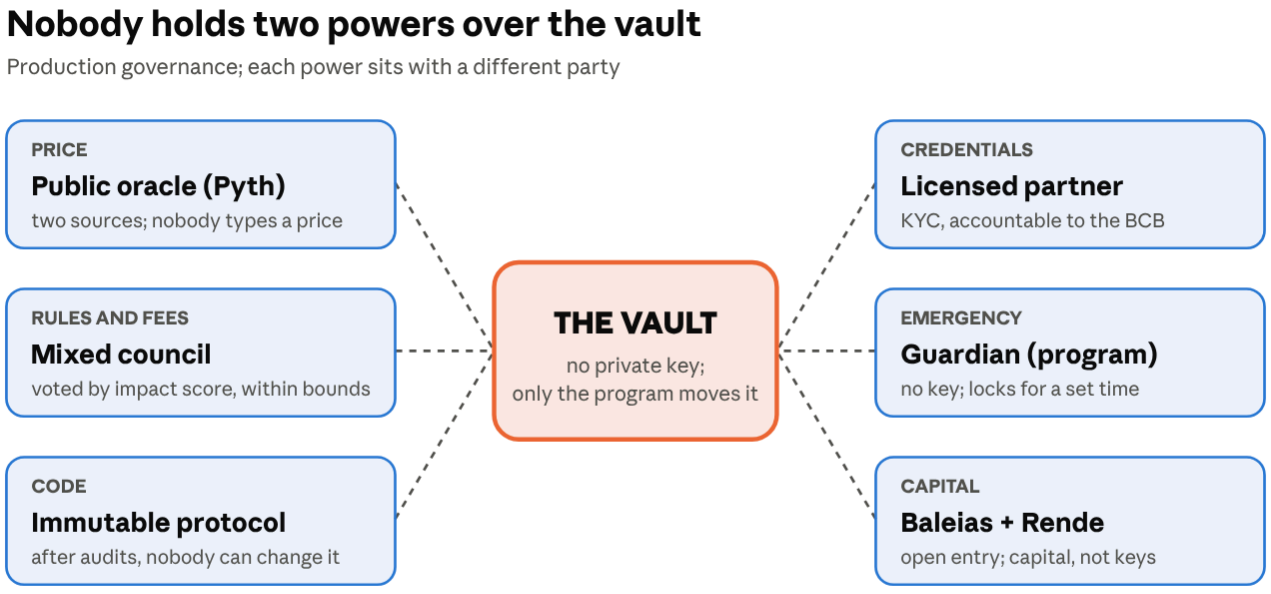
Risk	Safeguard in the program	How it was tested
Sanctions or court order	Asset freeze (Brazilian Law 13.810/2019): blocks swaps, deposits and withdrawals until released; revocation is separate	Integration test
Drain through a wrong price	Sliding-window breaker: at most 30% of each currency's liquidity leaves per hour, counting net outflow	Integration tests, including the round-trip attack
Bank run on the pool	Instant withdrawal up to 10% of liquidity; above that, a queue with advance notice and reserved funds	Integration test with a real wait
Abused emergency pause	There is no pause key: the guardian is a program that locks only by rule and for a set time, with a daily cap; lifting early belongs to the council	Devnet, 1,000 wallets: the deployer key tried to lock and was refused; a 5% price jump made the guardian rest on its own; the test council lifted it in 17 s
Run on one currency	Per-minute limit, hourly breaker and the guardian's breather on the draining side only	Stress: 19 of 40 R$ 1,000 buys went through, the rest were held; those returning dollars kept trading; the breather lifted on its own after 15 min
Rende currency short at withdrawal	30% of Rende's principal out of swaps; the queue accepts the request and pays in the other currency at the oracle price, if the person prefers	Integration test: reals short, a R$ 1,000 request paid in dollars at the oracle price, no fee
Malicious code upgrade	v2.5: governed upgrades with a timelock. v2.6 on devnet: the upgrade authority is still the deployer key (test only). v3: immutable code after audits; nobody can upgrade it	v2.5 applied on devnet through this path; early execution refused

Risk	Safeguard in the program	How it was tested
Unbacked currency	Proof of reserves: if issued supply exceeds the attested reserve, swaps and deposits stop	Integration test (off on devnet, no real custodian)
B2B flow out of control	Institutional per-operation limit and partner delisting	Integration test

An internal review of v2 found 8 issues, none critical, all fixed. An external audit is mandatory before real money.

Governance

No single party controls the price, the rules or the code. Each power sits with a different party, and every change waits in public before it takes effect.



production governance model; on devnet, a Squads v4 test multisig stands in for the council

- **Price:** a public, independent oracle (Pyth) with a second source; nobody types a price.
- **Rules and fees:** a mixed council (cambl, the licensed partner, an independent security firm, Rende depositors and Baleias, at most two seats each), voting by impact score within hard bounds written in code, with waits of 24 hours to 14 days.
- **Code:** the v3 protocol becomes immutable after two external audits: nobody can change it, not cambl, the council or the Baleias. cambl runs the front end (the app, the experience and the integrations), not the protocol. New features ship as a new version

that users choose to move to.
- **Credentials:** issued only by the licensed partner, accountable to the Central Bank.
- **Emergency:** a guardian that is a program, with no key: it locks for a set time (breathe, rest) or stops, by the rules written into it; it never lifts a lock or moves money; withdrawals keep working.

The full specification (golden rules, impact-score levels, voting chambers) is in Security architecture v3.

What runs on devnet today

Proven: the v2.5 governed upgrade. On Oct 16, 2026, the scarcity premium went live this way: proposal #2, 7 of 10 test approvals, early execution refused by the timelock, execution after 61 minutes, and the on-chain binary checked against the compiled one.

A finding on the way. Devnet refuses `ExtendProgramChecked` when called from inside a multisig (CPI), while `Upgrade` works. Since v2.5 is 3 KB larger, the extra program space is now added directly with the authority-free `ExtendProgram`, which only pays rent and changes no code, and the governed proposal carries only the `Upgrade`. We confirmed this with a disposable multisig and an empty test program before touching the real one.

An accident that confirmed the design. The v2.5 multisig's test keys lived only in the GitHub Actions cache, which deletes a cache after 7 days without use. Without the keys, nobody can update that pool's price or upgrade it any more: swaps and deposits are stuck, while principal withdrawals remain possible, because a withdrawal does not depend on the price, as the golden rule promises. That is why the guardian became a program, why the v2.6 test keys are derived from a single key, and why v3 is immutable: nothing day to day may depend on a key someone holds. The v2 branch app (not published yet) points to the v2.5 pool and will be migrated to v2.6; the public submission app stays on the original program, which was not touched.

Before mainnet: council seats go to the institutions named above, keys move to institutional MPC or HSM custody, and the guardian stays a program; people on call only receive the alerts and bring any early lifting of a lock to the council.

Security architecture v3: an untouchable vault

The next version removes every hand that could reach the money: no person, company or Baleia will be able to move customer funds outside the program's rules, and the protocol's code will become impossible to change.

The principle

The money must never depend on trusting anyone. The vaults already have no private key; v3 goes further, so that no single party controls the price, the rules or the code around them. On devnet, a test multisig plays the governance role to prove the mechanics (as with the v2.5 upgrade); it is scaffolding for the prototype, not the production design.

Separation of powers: nobody holds two

Power	Who holds it
Price (oracle)	A public, independent oracle (Pyth) with a confidence check and a second source; swaps stop if they disagree. Nobody types a price
Fees and limits	A mixed council, always within hard bounds in code, voted by impact score (below)
Program code	Nobody: the v3 code becomes immutable after audits. New features only as a new, opt-in version
KYC issuer	The licensed partner only, accountable to the Central Bank
Emergency pause	A keyless guardian program that locks for a set time (live on devnet, v2.6)
Providing liquidity and earning fees	Baleias: any level-3 investor, sized by profile, under a pool capacity cap

The mixed council gives each group at most two seats so that none holds a majority: cambl, the licensed partner, an independent security firm, a Rende depositor representative and Baleias. Its keys live in institutional MPC or HSM custody, spread across people and places, and rotated.

Baleias provide capital, not keys. They earn the junior share of fees and take first loss; price, fees, code and credentials are decided elsewhere. Open entry means there is no fixed club: as many Baleias as the pool's capacity and their investor profiles allow.

An immutable protocol

The next protocol cannot be changed. v3 is deployed, audited twice and then loses its upgrade authority for good (`Authority: none`), which anyone can check on any Solana explorer. From that moment nobody can alter its code: not cambl, not the council, not the Baleias, not an attacker who steals a key. cambl's job becomes the front end only: the app, the experience, customer support and the integrations with the partner and Pix.

Governance keeps only what the code itself allows: moving parameters within bounds written in code (fees, limits, the list of approved oracle feeds), by impact score and timelock. Every rule about money, including the bounds themselves, is fixed forever.

Golden rules in the protocol, impossible to switch off:

1. There is no "admin withdraws" instruction.
2. Each depositor can always withdraw their own principal, instantly or through the queue.
3. The vault never pays more than it holds; liabilities never exceed assets, checked on every operation.
4. Outflow per hour is capped (the circuit breaker becomes a fixed rule).

Because the code cannot change, improvements ship as a new protocol deployed alongside it. Nobody is moved by force: each user decides whether to migrate, by withdrawing from one and depositing in the other, and the golden rules guarantee that exit.

Governance by impact score

The program itself scores each proposal by its type and size; the proposer does not pick the score. The higher the impact, the more people vote and the longer everyone waits.

Level	Examples	Who approves	Wait
1. Low	Fee change up to 10% of its current value; limit within its band	Council, 3 of 5	24 hours
2. Medium	Larger fee change, new B2B partner, new KYC issuer	Council, 4 of 5; depositors can veto	72 hours
3. High	Oracle feed change (from the approved list), fee split change within bounds	Council, 4 of 5, plus depositor vote with quorum	7 days
4. Critical	Anything touching withdrawals or principal	All of the above, plus an exit window to withdraw first	14 days

Votes are counted in two chambers: Rende depositors vote one person, one vote, made sybil-proof by the KYC credential; Baleias vote by the square root of their capital, so 100 times the money gives only 10 times the vote. A change passes only if both chambers agree, with the council on top. Pending the legal opinion, depositors start with veto only rather than approval rights, to avoid strengthening a securities reading of Rende.

Defense in depth

- The guardian program: anyone calls the watch, and it locks for a set time when it sees a fast outflow, a price jump or missing backing.
- Proof of reserves, verifiable builds (`solana-verify`), and a bug bounty.
- An insurance fund built from a share of fees, to cover an exploit.
- Two independent audits before the code becomes immutable, and formal verification of the golden rules as a goal.

No system is unbreakable. The realistic goal is no single point of failure, a bounded worst case, and a guaranteed exit for every user.

Open where it builds trust, closed where it protects

The on-chain program stays open source with verifiable builds, under a Business Source License that allows reading and auditing but blocks commercial copies for a few years. The compliance engine (AML rules, fraud signals), partner and Pix integrations, risk models, and infrastructure stay private. Keys, credentials and customer data are never in code.

Order of delivery

1. Write the v3 specification and the golden-rule tests.
2. Build the v3 protocol with the golden rules as invariants; Pyth oracle; council and impact-score governance; open Baleia entry.
3. Run it on devnet with the 1,000-wallet scale test, as with v2.6.
4. Two external audits, then mainnet with small amounts and a bug bounty.
5. Remove the upgrade authority, making v3 immutable, and announce it publicly.

The current v2 program will not be frozen: it still has a broad admin and no external audit.

Compliance and regulation

The road to real money runs through a partner licensed by Brazil's Central Bank (BCB), with cambl as its technology. This is an informational reading, to be confirmed by a legal opinion.

Topic	Current reading	What cambI does
Virtual assets (Law 14.478/2022 and 2025 BCB rules)	Whoever operates the pool likely counts as a virtual asset service provider, needing prior authorization	Pilot as the technology of a licensed provider; admin keys under regulated governance
Foreign exchange (Law 14.286/2021)	Swapping a dollar-pegged stablecoin is an FX operation	Baleias are authorized institutions or qualified investors only (level-3 credential)
Securities (CVM Opinion 40)	Rende may be a public offering	In the pilot, Rende is for qualified investors only; no promised returns
Legal tender of the real	Obligations in Brazil are settled in reals	Paying a Pix QR with dollars: the merchant receives reals from the partner
AML/CFT and sanctions (BCB Circular 3.978, Law 13.810)	Monitoring, reporting to COAF, immediate asset freeze	Revocable credential, on-chain asset freeze, Travel Rule on transfers of US$ 1,000 or more
Data protection (LGPD, Constitutional Amendment 115/2022)	A wallet address is pseudonymous personal data	Nothing personal on-chain; the code↔person map stays with the partner
IOF tax	Rate on crypto operations still under debate	Configurable rate, shown separately from the fee

Three decisions come before launch: Rende's legal classification, backed by an opinion; the partner contract (who answers to the customer, IOF, COAF); and a formal data policy.

Measured results on devnet

A thousand test wallets ran the v2.6 pool end to end on Oct 28, 2026: sign-up with a credential, the KYC gate, swaps in both directions, Rende, the withdrawal queue, the keyless guardian and an economic stress test. All 27 claims in this paper checked out, and all 4,547 transactions gave the expected result, in 76 minutes (the pace is the free public RPC's limit, not Solana's).

deployments/load-v2-report.json (scale test with 1,000 wallets), v2 branch, Oct…

The finding that changed the product. In the first test (Oct 16, 8 wallets), buying dollars cost **1.2%**; bringing dollars cost **0.33%**. The pool started with about 20% more reals than dollars, and the dynamic fee did its job: it charged whoever worsened the imbalance and rewarded whoever helped. That is too expensive for retail, which led to three fixes:

1. **Scarcity premium on deposits (live; with 1,000 wallets, +2.23% shares for those bringing dollars).** Depositing the scarce currency earns up to 10% more shares, in proportion to the imbalance, with no premium if the deposit overshoots.
2. **A balanced pool at launch,** seeded 50/50 by the Baleias. The 1,000-wallet test showed why: the Baleias came in 50/50, but Rende came in reals only and left swap liquidity with 26% extra reals.
3. **A retail base fee of 0.6% to 0.8%,** to be set by governance with pilot data.

In the 1,000 wallets' normal trading, the breaker peaked at 4% of its limit in reals and 5.4% in dollars; in the stress test's dollar run, at 68%. There were no unexpected refusals: every one was planned (no credential, above the limit, deposit lockup, per-minute limit, the guardian's breather and rest).

Economic stress test

At the end of the test, 40 wallets tried to take R$ 1,000 out in dollars each, in bursts, and then all of them brought the dollars back. What happened:

- **The locks held the run.** In the first burst, 19 buys went through and the per-minute limit held 21. At the next watch, the guardian ordered a breather on dollars only, and the second burst was refused in full by the breather.
- **The fee rose as dollars ran short:** from 1.22% on the first buys to 1.33% on the last.
- **Those bringing the scarce currency kept trading** during the breather and paid a 0.15% median. Nobody profited on a round trip (0 of 19).
- **The pool came back to where it started:** imbalance went from 26% to a 40% peak and back to 26%.
- **The breather lifted on its own** after 15 minutes, with nobody touching it.

An honest detail: on one return trade, the rebalancing discount put the fee at −0.01% (the person got slightly more than the oracle price). Even so, the round trip was not profitable, which is the guarantee written in the program.

Go-to-market and traction

We start with Brazilians who earn in dollars, because they convert every month, feel the fee, and already search for alternatives.

Beachhead: freelancers and remote workers paid from abroad, then small exporters of digital services. **Expansion:** partner apps (B2B) that embed cambI as their FX engine, and tourists paying Pix QR codes with dollars.

North-star metric: weekly active wallets, meaning at least one swap or payment in 7 days. It is measured on-chain by the public indexer, with aggregates only (the Oct 28, 2026 scale test used 1,000 wallets; the indexer still reads the v2.5 pool and will be pointed at v2.6).

Channel	Why it works	First experiment
Landing page for freelancers	They compare fees and switch for a cheaper option	"How much did you lose on your last transfer?" calculator plus a waitlist
Shareable savings receipt	Every receipt shows the saving versus a bank, which turns users into a channel	Already in the app; measure shares and sign-ups
Communities	Brazilian dev and freelancer groups, Superteam, Solana Brasil	Posts with real numbers, no promise of returns
SME exporters	Recurring volume, B2B fee	5 pilot conversations through a partner brokerage
Partner apps (B2B)	One integration brings many users without per-person KYC cost	Letters of intent from 2 apps that pay freelancers
Tourists	Pay any Pix QR with dollars	Pilot in one tourist area once the partner is live

90-day targets

Target	Number	Why it matters
Waitlist sign-ups	500	Proves demand in our own channel
Customer interviews (monthly volume and current cost)	50	Real data on what users pay today

Target	Number	Why it matters
Licensed partner in negotiation	1	Without it there is no real money
Letters of intent from B2B apps	2	Volume beyond retail
External audit quoted and scheduled	1	Required before mainnet

These are our goals, not results. Each one will be reported in public Builder Updates.

Roadmap

cambl roadmap, October 2026

Stage	Delivery	Gate to move on
Done	v1 (submission); v2.4 to v2.6 on devnet: on-chain KYC, sliding breaker, withdrawal queue, scarcity premium through a governed upgrade, keyless guardian, Rende reserve and a 1,000-wallet test	27 of 27 claims confirmed with 1,000 wallets; v2.5 executed after the 1-hour wait
Now	Legal opinion on Rende; model contract with a partner; retail base fee set by governance	Opinion and contract signed
Pilot	One licensed brokerage as KYC issuer and Baleia; Rende for qualified investors only; external audit	Weekly active wallets growing; zero incidents
Niche	Mainnet with freelancers and exporting SMEs; yield-bearing backing (tokenized Selic bonds and T-bills)	Volume that sustains the Baleias
Scale	B2B apps, QR payments for tourists, capacity cap with an entry queue	Competitive retail fee without subsidies

Security v3 (immutable protocol, separation of powers, impact-score governance, open Baleia entry) is built between Now and Pilot; the v3 protocol becomes immutable only after two external audits, before real money at scale.

Team and hiring

cambl was designed and built by its founder, Gustavo: the program, the app, the governance setup and the measurement pipeline. We are now looking for people to take it from devnet to real money.

Role	Focus	When
Technical lead, Solana and TypeScript	Program, app, infrastructure and security through mainnet	Now
Regulatory counsel (BCB, CVM)	Rende's classification, virtual-asset framework, partner contract	Now
Business development	Licensed partner and the first Baleias	Now
Compliance / AML lead	KYC/KYB policy, monitoring, COAF reporting through the partner	Pilot
External smart contract audit	Full review of the v2.6 program and the guardian before real funds	Before mainnet
Product designer	Onboarding, accessibility, usability tests with freelancers and tourists	Pilot
Growth and content	30-day plan, waitlist, community	Pilot
Accounting and finance	Company setup, taxes, IOF with the partner	Pilot
Data protection officer	Data policy, impact report, user rights channel	Pilot
Data engineer	On-chain indexer, metrics, compliance alerts	After the pilot

Regulated roles (legal, accounting) require licensed professionals.

Risks and disclaimer

Risk	What could happen	Mitigation
Oracle	A stale quote becomes profit for arbitrageurs and a loss for depositors (simulation: ~12% of the pool per year with hourly updates)	Deviation-triggered updates (~1 min), two sources agreeing within 0.3%, a cap on change per update

Risk	What could happen	Mitigation
One-way flow	Brazilians buy dollars and the dollar side dries up	Dynamic fee, rebalancing discount, scarcity premium, Baleias restocking dollars
FX risk	The Baleia holds the open position between real and dollar	Baleias are junior and professional; an open-position cap on the roadmap
Smart contract	A bug in the code	Tests, internal review, external audit before real money, immutable code after audits (v3)
Regulation	Rule changes, IOF tax, Rende's classification	Licensed partner, restricted Rende, configurable rates
Adoption	Low volume dilutes depositors' returns	Capacity cap, focus on B2B and inbound payments from abroad

Disclaimer. This document describes a prototype on a test network, using tokens with no real value. It is not an investment offer or a promise of returns. Returns depend on swap volume and can be negative for Baleias; deposits are not covered by Brazil's deposit guarantee fund (FGC). The legal and regulatory analysis here is informational and needs an opinion from licensed professionals. Market figures are estimates.

Technical appendix

Everything below is taken from `programs/cambi_pool/src/lib.rs` (v2.6, branch `v2`) and, for the guardian, from `programs/cambi_guardian/src/lib.rs`.

To prove the mechanics before the council exists, a test council (a Squads v4 multisig, 7 of 10 test keys) holds the admin and oracle roles of the v2.6 pool (`8PxeD9tYtnEhaY6x59sEirKeFdrCP3Bx3yEyThU1RiFQ`). The test keys are derived from the deployer key, so none depends on its own cache. The guardian is the `cambi_guardian` program: nobody holds a pause key. In the v2.5 pool, one test multisig held the settings and another the upgrade authority, behind a 1-hour timelock.

Pool imbalance

With B the free reals, U the free dollars and p the oracle price (reals per dollar):

```
I = \frac{B - U p}{B + U p}, \qquad -1 \le I \le 1
```

I > 0 means the pool holds extra reals (dollars are scarce).

Dynamic fee

A swap that worsens the imbalance pays more; one that improves it pays less. With w = I when reals come in and w = −I when dollars come in:

$$\text{fee} = \text{base} \cdot \operatorname{clamp}(1 + w,\ 0.7,\ 1.5)$$

Base fees: retail 1.0%, depositors 0.5%, B2B 0.4%.

Rebalancing discount

Whoever brings the scarce currency gets extra output, computed on the imbalance left after the swap (I_2), with k = 2%, only if the swap does not overshoot (I_2 keeps the sign of I):

$$\text{bonus} = \min\left(k \cdot |I_2|,\ \text{fee} + \text{base}\right)$$

The cap guarantees a round trip never profits.

Scarcity premium on deposits (v2.5)

Depositing the scarce currency mints extra shares, again on the imbalance left after the deposit, with no premium on overshoot:

$$\text{premium} = \min\left(5k \cdot |I_2|,\ 10\%\right), \qquad \text{shares} = V_{\text{BRL}} \cdot (1 + \text{premium})$$

Principal stays nominal: the premium only raises the depositor's slice of future fees.

Circuit breaker

Net outflow of each currency in a sliding one-hour window is capped at 30% of its base. Past outflow decays linearly; inflow of the same currency offsets it:

$$\text{used}(t) = \text{used}(t_0) \cdot \max\left(0,\ 1 - \frac{t - t_0}{W}\right), \qquad \text{used} \le 30\% \cdot \text{base}$$

Limits and withdrawals

Rule	Level 1 (basic)	Level 2 (full)	Level 3 (institutional)
Per operation	R$ 10,000	R$ 100,000	no cap
Per 24 hours	R$ 30,000	R$ 300,000	no cap

Rule	Level 1 (basic)	Level 2 (full)	Level 3 (institutional)
Baleia deposits	no	no	yes

Instant withdrawal up to 10% of free liquidity; above that, a withdrawal ticket with advance notice (minimum 60 s; 10 min on devnet), the amount reserved in `queued`.

Rende reserve: swaps only use the liquidity left after keeping 30% of Rende's principal in each currency (the council can set it between 20% and 100%). If a currency's queue exceeds what is in the vault, swaps that take that currency stop until the queue is covered; at the end of the notice, the person claims in the requested currency or, only while it is short, receives the same value in the other currency, at the oracle price and with no fee.

Guardian and locks (v2.6)

At every watch, the guardian compares the pool with the previous snapshot. The most serious level wins:

Level	Trigger	Lock	How long
1. Breathe	Swap liquidity in one currency fell 20% or more within a 15-minute window	Only swaps that take that currency	15 min (pool cap: 30 min)
2. Rest	Price jumped 3% or more between two watches, or 3 breathers in one day	Swaps and deposits	1 h (cap: 2 h at a time, 6 h a day)
3. Stop	The vault does not cover reserved fees and the Rende reserve, or Baleia equity falls below 10% of what is owed to Rende	Full pause	until the council lifts it

Withdrawals are never blocked. Only the council can lift a lock early.

Accounts

Account	Address seeds	Holds
Pool	`pool`, real mint, dollar mint	Price, fees, limits, breaker state, guardian locks, Rende reserve, governance keys

Account	Address seeds	Holds
Vault (one per currency)	`vault`, pool, mint	The money; owned by the pool
Credential	`credential`, pool, user	Level, expiry, issuer, revoked, frozen, daily usage
Position	per user and layer	Principal, shares, reward debt
Withdrawal ticket	`ticket`, position	Queued amount and ready time
Reserve attestation	`reserve`, pool, mint	Attested reserve, time, attester
Watch (guardian program)	`watch`, pool	Last watch snapshot, 15-minute window, strikes of the day

Tests

- 34 Rust unit tests: 27 in the pool (fees, imbalance, premium, breaker, limits, timed locks, Rende reserve) and 7 in the guardian (breathe, rest and stop decisions).
- Integration tests on a local Solana in CI for every rule above, including the attacks (round trip, breaker inflation, griefing).
- 101 app tests checking that the instructions the app builds match the published IDL.
- Scale test on devnet (v2.6): 1,000 wallets, 4,547 transactions, 27 of 27 claims confirmed (earlier, on v2.5: 33 of 33 steps with 8 users).
- The CI checks that the production binary contains no devnet-only instructions.

References

cambi (branch `v2` of github.com/powerdeblack/cambi)

- Program: `programs/cambi_pool/src/lib.rs`; ecosystem report: `deployments/ecosystem-v2-report.json`; metrics: `deployments/metrics-v2.json`; governed upgrade history: `deployments/devnet-v2.json`; CPI diagnosis: `deployments/diag-loader-cpi.json`; guardian: programs/cambi_guardian/src/lib.rs; v2.6 record: deployments/devnet-v26.json; scale test: deployments/load-v2-report.json.

Market

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- 90% of crypto flow in stablecoins (BCB governor, Feb 2025): Reuters via Investing.com.

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- Stablecoin payment costs by region: Cointelegraph post, Oct 16, 2026.

Competitors and taxes

- Wise: Exiap, Euro Dicas. Nomad: Corretora Forex, Melhores Cartões. Remessa Online: Remessa Online, Wise blog. Banks: Exiap.
- IOF: o-tributo, Barbieri Advogados, Remessa Online; proposed IOF on crypto: ASA.

Costs

- KYC reference price: Didit. Pix fees for businesses: Contabilizei. SOL price: MetaMask.

Web figures were collected on Oct 17, 2026 from search summaries; several sources disagree, and each will be confirmed against the primary source before external use.